Scalability & Future Plan

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local JSON file storage used for sensor logging instead of a centralized cloud database.	Concurrent multi-regional telemetry inputs will lead to file I/O locks and performance degradation.	**High**
2	ResNet50 model leaf disease diagnostic inferences run synchronously on a single CPU device endpoint.	Simultaneous image uploads from multiple farmers will cause latency overheads and API timeout faults.	**Medium**
3	Weather notification alerts rely on periodic API polling routes instead of event-driven streaming channels.	Critical agricultural weather warnings might be delayed up to an hour depending on interval polling triggers.	**Low**

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles single local user session smoothly via local dev server.	Deploy application containerized using Docker containers behind an Nginx load balancer to distribute user traffic evenly.
2	Data Storage	Local dynamic tracking text storage files.	Migrate historical engine yields and field metrics to a hosted PostgreSQL instances with Redis indexing configurations.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
3	Performance	Synchronous ML execution sequences.	Offload computer vision processing routines onto asynchronous Celery worker nodes backed by GPU inference instances.
4	Security	Basic hardcoded endpoint validation keys.	Implement OAuth2 validation pathways alongside strict token-based request encryption configurations.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud Migration & Multi-region DB Setup	Month 1-2	Enables dynamic data synchronization from vast agricultural sectors seamlessly.
Phase 2	Offline-first Mobile Deployment Core	Month 3-4	Provides continuous analytics monitoring to remote regions with weak cellular connectivity.
Phase 3	Automated Droning Telemetry Interoperability	Month 5-6	Allows aerial image gathering to bypass manual crop verification tasks entirely.
Phase 4	B2B Fertilizer Market Integration Platform	Month 7-8	Directly bridges targeted soil recommendation values with accessible localized suppliers.